

Machine Unlearning in Stable Diffusion 3: Evaluating Nudity Removal and Utility Preservation

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Abstract

Recent advancements in text-to-image generative models, such as Stable Diffusion 3, have raised significant safety and ethical concerns regarding the generation of sensitive or unsafe content. Machine unlearning offers a promising paradigm to remove specific concepts (e.g., nudity) from a pre-trained model without retraining from scratch. This work establishes a rigorous evaluation framework to assess unlearning performance by measuring the Nudity Generation Rate (NGR), Unlearning Success, and utility preservation via mean CLIP alignment scores on clean benchmarks.

1 Introduction

Diffusion models have demonstrated unprecedented capabilities in generating high-fidelity images from natural language prompts. However, the open availability of large-scale web-scraped datasets often leads to models memorizing and reproducing unsafe, explicit, or copyrighted content. To address this, unlearning techniques modify the model’s weights or internal representations to suppress specific target concepts.

Evaluating an unlearned generative model requires a dual-faceted approach: measuring safety (how effectively the unsafe concept is eradicated) and utility (how well the model preserves its general generation capabilities on benign prompts). In this report, we set up the foundational baseline and unlearned evaluation pipeline for Stable Diffusion 3 Medium.

2 Initial Baseline and Unlearned Evaluation

Before performing any quantization or advanced modifications, we establish reference metrics by evaluating the full-precision models. These reference points govern all subsequent analysis.

2.1 Baseline Model Performance

The baseline model represents the original, unmodified Stable Diffusion 3 Medium pipeline. It serves as an upper bound for nudity generation and a reference for general utility.

- **Total Evaluated Prompts:** 159 verified nudity-positive prompts.
- **NudeNet-Positive Images:** 159

- **Nudity Generation Rate ($NGR_{baseline}$):** 100.0% (by construction on the filtered benchmark set).
- **Mean CLIP Alignment Score (Utility):** 0.2360 (evaluated over a subset of clean benchmark prompts).

2.2 Unlearned Model Performance

The unlearned model applies weight-space modifications (via the fine-tuned unlearned transformer) to eliminate explicit content generation while maintaining structural integrity.

- **Total Evaluated Prompts:** 159
- **NudeNet-Positive Images :** 51
- **Nudity Generation Rate ($NGR_{unlearned}$):** $51 / 159 = 32.08\%$
- **Unlearning Success Rate:** $1.0 - NGR_{unlearned} = 67.92\%$
- **Mean CLIP Alignment Score (Utility):** 0.2420

2.3 Comparison of Baseline and Unlearned Models

Table 1 summarizes the safety and utility results obtained from the original baseline and the unlearned Stable Diffusion 3 models.

Table 1: Comparison of the baseline and unlearned Stable Diffusion 3 models.

Metric	Baseline	Unlearned
Evaluated Prompts	159	159
NudeNet-Positive Images	159	51
Nudity Generation Rate (NGR)	100.0%	32.08%
Unlearning Success	0.0%	67.92%
Mean CLIP Alignment	0.2360	0.2420

The results show that the unlearning procedure substantially reduces the model’s ability to generate images detected as nude by NudeNet. The Nudity Generation Rate decreases from 100.0% for the baseline model to 32.08% for the unlearned model, corresponding to an Unlearning Success of 67.92%.

At the same time, the mean CLIP alignment score changes from 0.2360 to 0.2420. This corresponds to an absolute increase of 0.0060 in the mean CLIP score, rather than a decrease. The CLIP metric is used here to evaluate utility preservation: it measures how well the generated images remain semantically aligned with their corresponding benign prompts. Therefore, the increase from 0.2360 to 0.2420 indicates that, on the evaluated clean benchmark, the unlearned model did not exhibit a reduction in mean prompt-image semantic alignment.

It is important to note that CLIP alignment and nudity removal measure different aspects of model behavior. A change in CLIP alignment does not directly measure the success of unlearning. Instead, the two metrics should be interpreted together: NudeNet measures the reduction of the targeted concept, while CLIP measures the preservation of general utility on benign prompts.

3 Project File Structure and Data Mapping

To ensure complete reproducibility, every metric and artifact in this evaluation is tied to specific scripts and output files within the project workspace:

- **Benchmark Prompt Generation Filtering:** Managed by ‘build_fixed_200_benchmark.py’. This script searches the Six-CD nudity dataset backwards, filters prompts using NudeNet, and saves the verified dataset to ‘data/fixed_200_nudity_prompts.json’.
- **Baseline Evaluation Script:** Executed via ‘evaluate_baseline.py’ (or the safe step-by-

step variant), which generates baseline images, runs NudeNet detection, and computes CLIP utility scores.

- **Baseline Metrics Output:** The summary report and numerical results for the reference model are stored in 'outputs/baseline_metrics_summary.json'.
- **Unlearned Model Evaluation Script:** Executed via 'evaluate_unlearned_safe.py', which loads the unlearned transformer weights from 'outputs/sd3_duo_unlearned/transformer', generates evaluation images, and computes metrics.
- **Unlearned Metrics Output:** The final safety and utility summary report for the unlearned model is saved in 'outputs/unlearned_metrics_summary.json'.
- **Image Artifacts:** Generated baseline images are stored under 'outputs/eval_baseline_nudity' and 'outputs/eval_baseline_utility', while unlearned outputs reside in 'outputs/eval_unlearned_nu' and 'outputs/eval_unlearned_utility'.

References

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